COURSE GUIDE & SYLLABUS · GRADES 6–8 · AGES 12–14

Scratch (Gr 6–8) Advanced: Algorithms Lab

Search, sort, encrypt and recurse — then carry it all across to Python.

The final middle-school Scratch course. Learners implement and measure classic algorithms — linear and binary search, swapping and bubble sort, string processing and the Caesar cipher, and recursion with fractals — then translate their thinking into Python. Projects: Search Showdown, Live Leaderboard, Secret Messenger and a capstone Algorithm App with a Python version. Every lesson has a video, build-along, auto-checked Code Lab challenges (tested with sample inputs), a worksheet and a teacher lesson plan.

What learners will be able to do

- Implement linear and binary search and handle “not found”
- Measure algorithms by counting steps and compare their efficiency
- Swap safely with a temporary variable and implement bubble sort
- Describe linear, logarithmic and quadratic growth informally
- Process strings character by character (count, reverse, palindromes)
- Encrypt and decrypt with a Caesar cipher using modular arithmetic
- Write recursive procedures with a base case, including fractals
- Translate between Scratch and Python (loops, conditionals, lists, functions)
- Design an algorithm in pseudocode, implement it, test it and present it

How every lesson works

Watch	A short animated explainer video with captions (and an optional read-aloud).
Learn	Learn cards with the real coding blocks, and a step-by-step build-along.
Try it (Code Lab)	Code in the built-in Scratch-style Code Lab with a live stage. Challenges are checked automatically and give a clear next step; hints unlock one at a time.
Puzzles & quick checks	Robot puzzles and questions that build understanding, with instant feedback.
Probe	Debugging and “new situation” tasks to show real understanding.
Projects	Each module ends with a creative project built in the Code Lab and reviewed by the teacher with a rubric.
Printables	A worksheet for every lesson, a project brief for every project, and a lesson plan with answer keys for the teacher.

Scope and sequence

#	Lesson	Big idea	Learners build

1	**Linear Search** Implement linear search to find an item in a list, report its position or “not found”, and count the steps it takes.	Searching algorithms	Find a friend
2	**Binary Search: Divide and Conquer** Implement binary search on a sorted list using low, high and mid, and compare its efficiency with linear search.	Binary search	Binary search 1–100
3	**Project: Search Showdown** Build an experiment that runs linear and binary search on the same data, counts their steps, and reports which is more efficient.	Project: comparing algorithms	Project: Search Showdown
4	**Swapping and the Bubble Pass** Swap two list items safely with a temporary variable, and write one bubble-sort pass that moves the largest value to the end.	Swapping and comparisons	Swap, then one pass
5	**Bubble Sort and Efficiency** Complete bubble sort with nested loops, sort in either order, and measure how the number of comparisons grows with the list size.	Sorting and complexity	Full bubble sort
6	**Project: Live Leaderboard** Plan, build and test a leaderboard app that collects scores, sorts them and shows the top three.	Project: sorting app	Project: Live Leaderboard
7	**Strings: Working with Text** Treat text as a sequence of characters: loop through letters, count matches, reverse a word and test for palindromes.	String processing	Vowel counter and reverser
8	**Cryptography: The Caesar Cipher** Encrypt and decrypt messages with a Caesar cipher using letter positions and modular arithmetic.	Ciphers and modular arithmetic	Encrypt a word
9	**Project: Secret Messenger** Plan, build and test an app that encrypts and decrypts messages with a chosen key, organised into procedures.	Project: cryptography app	Project: Secret Messenger
10	**Recursion: Blocks That Call Themselves** Write recursive procedures with a base case and a recursive case, and use recursion to draw a fractal tree.	Recursion	Countdown and fractal tree
11	**From Blocks to Python** Translate between Scratch blocks and Python: variables, loops, conditionals, lists and functions — and implement a classic algorithm from Python code.	Bridging to text code	FizzBuzz from Python
12	**Capstone: Algorithm App** Design, build, test and present an app built around an algorithm (searching, sorting, ciphers or recursion), with the matching Python written alongside.	Capstone project	Project: Algorithm App

Modules

Module 1 · Searching
Linear and binary search, step counting and efficiency. Project: Search Showdown.

Big question: How do computers find things so fast?

Module 2 · Sorting
Swapping, bubble passes, bubble sort and quadratic growth. Project: Live Leaderboard.

Big question: What does it cost to put things in order?

Module 3 · Strings and secrets
Character processing, palindromes and the Caesar cipher. Project: Secret Messenger.

Big question: How can code read, change and hide text?

Module 4 · Recursion and Python
Recursion and fractals, translating to Python, then a capstone Algorithm App.

Big question: How do the big ideas carry over to text code?

Assessment

Coding challenges are checked on the server by running the learner's program. Quick checks give instant feedback. Written answers and the four projects are reviewed by the teacher against 4-level rubrics (Beginning · Developing · Secure · Excellent).

What you need

- Laptop or Chromebook with a modern browser and a keyboard
- The built-in Kitabash Code Lab (no account needed)
- Optional: Scratch (scratch.mit.edu) and a free online Python editor

Before you start

- Scratch (Gr 6–8) Intermediate: Data & Simulations, or confident use of lists, procedures with inputs and validation

Safety and privacy

Cipher lessons are for fun and learning; remind learners never to use made-up ciphers to protect real passwords or personal data. Sharing to scratch.mit.edu is optional and needs a parent-approved account for under-13s. Learners under 18 use the AI tutor only with guardian consent.

What comes next

After this course: **Python (Gr 6–8) Beginner**, which goes deeper into clones, custom blocks, lists and bigger games.

Kitabash original curriculum. Scratch is a project of the Scratch Foundation in collaboration with the MIT Media Lab; this course is not affiliated with or endorsed by them.